

Predicting
Disasters Before
They Strike

AI-powered flood, earthquake, and wildfire prediction system for Pakistan. Real-time monitoring, risk assessment, and early warnings to save lives.

Get StartedWatch Demo

0.82

Flood Risk



12 Alerts

Active Now



Comprehensive Disaster Protection

Advanced AI models analyze satellite data, weather patterns, and seismic activity to predict disasters before they occur.

Flood Prediction
LSTM-based rainfall analysis and river level monitoring for accurate flood forecasting with 72-hour advance warnings.

- ✓ Rainfall prediction
- ✓ River monitoring
- ✓ Risk zone mapping

Earthquake Prediction
Seismic activity analysis using USGS data and geostrophic patterns to identify high-risk zones across Pakistan.

- ✓ Seismic monitoring
- ✓ Fault line analysis
- ✓ Intensity prediction

Wildfire Detection
Satellite-based fire hotspot detection using MODIS VIIRS and thermal anomaly detection for rapid response.

- ✓ Satellite monitoring
- ✓ Hotspot detection
- ✓ Ground prediction

Powerful Dashboard Interface

Real-time risk assessment with interactive maps, detailed analytics, and instant alerts.



How It Works

Four simple steps to comprehensive disaster protection

1

Data Collection
Real-time data from NASA, USGS, and OpenWeather APIs

2

AI Analysis
LSTM models process data and predict risk scores

3

Risk Assessment
Interactive maps show danger zones by city

4

Early Alerts
Instant notifications when risk thresholds cross

Ready to Protect Your Community?
Join thousands of users who trust DisasterGuard for early warning and disaster preparedness.

Create Free Account



Welcome Back

Sign in to access your disaster monitoring dashboard

 Google

 GitHub

or continue with email

 Enter your email

 Enter your password

Remember
me

[Forgot password?](#)

 Sign In

Don't have an account? [Create one](#)

Create Account

Join DisasterGuard and stay protected

 Google

 GitHub

or register with email

 First Name

 Last Name

 Enter your email

 Phone number (optional)

 Create password

 Confirm password

☐ I agree to the [Terms of Service](#) and [Privacy Policy](#)

 Create Account

Already have an account? [Sign in](#)

MAIN

Dashboard

Alerts

History

Reports

DISASTER MONITORING

Flood

Earthquake

Wildfire

Risk Heatmap

SETTINGS

Preferences

Dashboard

Welcome back! Here's your disaster monitoring overview.

Refresh

Export Report

Flood Risk

HIGH

0.73

Karachi, Sindh

View Details

Earthquake Risk

MODERATE

0.45

Islamabad Region

View Details

Wildfire Risk

LOW

0.12

Punjab Region

View Details

Risk Heatmap - Pakistan

Full View



Low (0-0.3) Moderate (0.3-0.6) High (0.6-0.8) Extreme (0.8+)

Recent Alerts

View All

Flood Risk Increased

2 min ago

Karachi - Heavy rainfall predicted

Earthquake Detected

15 min ago

Magnitude 4.2 near Islamabad



12

Cities Monitored



847

API Calls Today



3

Active Alerts



95%

System Uptime

Recent Activity

View All

Event Type	Location	Risk Level	Time	Status	Action
Flood	Karachi, Sindh	HIGH (0.73)	2025-04-16 14:30	Monitoring	Details
Earthquake	Islamabad	MOD (0.45)	2025-04-16 12:15	Active	Details
Wildfire	Punjab Region	LOW (0.12)	2025-04-16 08:00	Normal	Details
Heavy Rain	Lahore, Punjab	MOD (0.58)	2025-04-15 22:45	Alert	Details

DisasterGuard

Back to DashboardJD

MAIN

DashboardAlertsHistory

MONITORING

FloodEarthquakeWildfireHeatmap

🔔 All Alerts

Manage your notifications and warnings

🔴 3Critical Alerts

⚠️ 5Warning Alerts

ℹ️ 12Info Alerts

Alert History

Flash Flood Warning

Malir River overflow expected in Karachi

2 min ago

Earthquake Detected

Magnitude 4.2 near Islamabad

15 min ago

Fire Hotspot Detected

Satellite data shows activity near Punjab

1 hour ago

DisasterGuard

Back to DashboardJD

MAIN

DashboardAlertsHistory

MONITORING

FloodEarthquakeWildfireHeatmap

🕒 Historical Data

View past events and download reports

mm/dd/yyyy

to

mm/dd/yyyy

All Types

Export CSV

DATE/TIME	EVENT TYPE	LOCATION	SEVERITY	DETAILS	ACTIONS
2025-04-16 14:30	Flood	Karachi, Sindh	HIGH (0.73)	Heavy rainfall 145mm	View
2025-04-16 12:15	Earthquake	Islamabad	MOD (0.45)	Magnitude 4.2, Depth 12km	View
2025-04-15 22:45	Flood	Lahore, Punjab	MOD (0.58)	89mm rainfall, minor flooding	View

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MONITORING

Flood

Earthquake

Wildfire

Flood Monitoring

Real-time rainfall analysis and flood risk assessment

Karachi, Sindh

Refresh

Current Risk Score

0.73

HIGH RISK

Risk Breakdown

Rainfall Intensity 85%

River Water Level 72%

Soil Saturation 60%

72-Hour Forecast

120mm 150mm 45mm

LSTM Rainfall Prediction



Current Metrics

Current Rainfall

45.2 mm

River Level

8.4 m

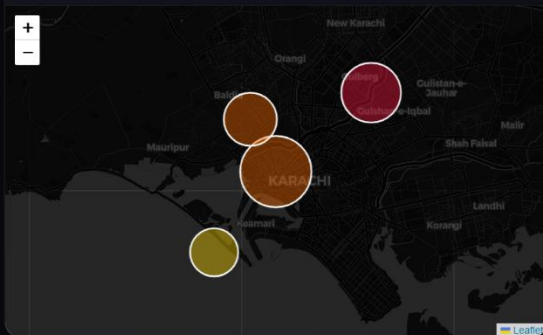
Soil Saturation

78%

Drainage Capacity

62%

Flood Risk Zones



Active Warnings

Flash Flood Warning

Malir River overflow expected

Heavy Rain Alert

150mm+ rainfall predicted

Road Closure

Shahrah-e-Faisal flooded

Flood History

DATE	LOCATION	RAINFALL	RISK SCORE	STATUS	ACTIONS
2025-04-16	Karachi, Sindh	145mm	0.73	Active	View
2025-04-10	Lahore, Punjab	89mm	0.52	Resolved	View
2025-04-05	Multan, Punjab	67mm	0.28	Normal	View

- MAIN
- Dashboard

Alerts

History
- MONITORING
- Flood

Earthquake

Wildfire

Earthquake Monitoring

Karachi, Sindh

Refresh

Real-time seismic activity analysis and prediction

Seismic Risk Score

0.45

MODERATE

Risk Factors

Magnitude Activity

4.2 ML

Fault Line Proximity

12 km

Historical Activity

Moderate

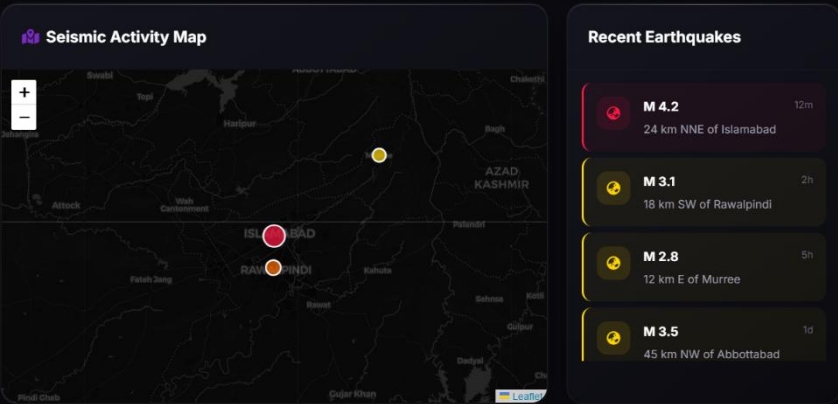
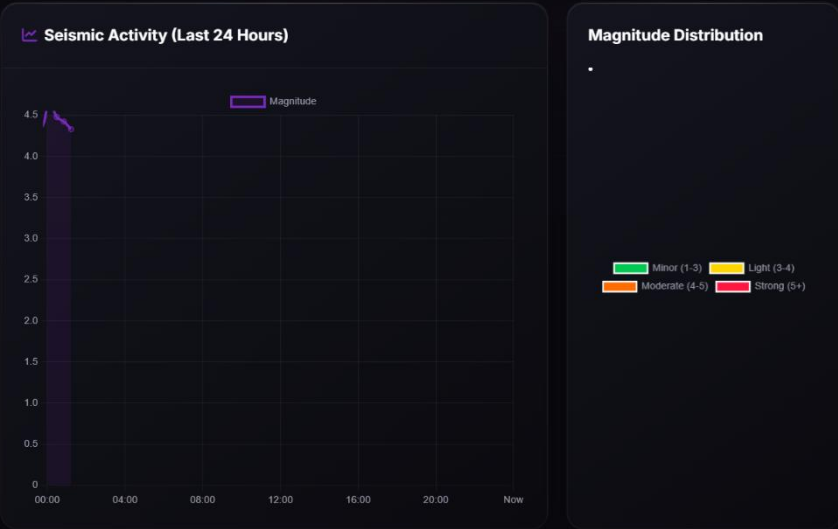
Recent Events

M 4.2

12 min ago

M 3.1

2 hours ago



Earthquake History

DATE/TIME	MAGNITUDE	LOCATION	DEPTH	INTENSITY	STATUS
2025-04-16 14:30	4.2 ML	Islamabad Region	12 km	Moderate	Recent
2025-04-16 12:15	3.1 ML	Rawalpindi	8 km	Light	Recorded
2025-04-15 09:45	2.8 ML	Murree	15 km	Minor	Recorded

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Earthquake

Wildfire

Wildfire Monitoring

Satellite-based fire hotspot detection and analysis

Punjab Region

Refresh

Wildfire Risk Score

0.12

LOW RISK

Risk Factors

Temperature

38°C

Humidity

45%

Wind Speed

15 km/h

Satellite Status

Active Hotspots

0 detected

Last Scan

15 min ago

Temperature Trend (7 Days)



Fire Risk Conditions

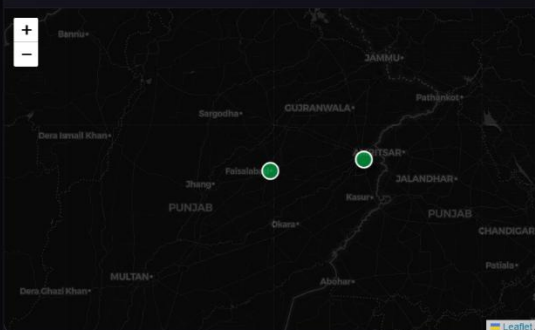
Normal Conditions

No immediate fire risk

Moderate Heat

Monitor conditions

NASA FIRMS Fire Map



Satellite Data Sources

VIIRS SNPP

375m resolution

Active

MODIS Terra/Aqua

1km resolution

Active

GOES-16

Geostationary

Active

Fire Detection History

DATE	LOCATION	BRIGHTNESS	CONFIDENCE	SATELLITE	STATUS
No active fire hotspots detected All clear in monitored regions					

MAIN

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MONITORING

[Flood](#)[Earthquake](#)[Wildfire](#)[Heatmap](#)

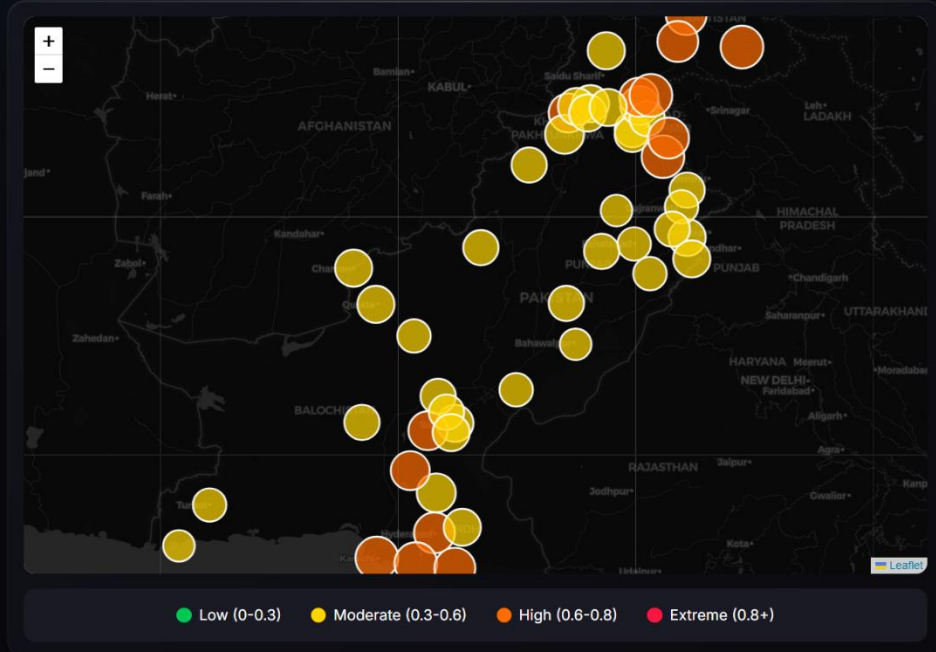
Interactive Risk Heatmap

Search city...

All Risks

Refresh

View disaster risk zones across Pakistan



MAIN

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MONITORING

[Flood](#)[Earthquake](#)[Wildfire](#)[Heatmap](#)

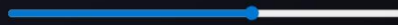
Settings

Manage your preferences and alert thresholds

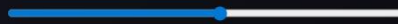
Save Changes

Alert Thresholds

Flood Risk Threshold 0.60



Earthquake Risk Threshold 0.50



Wildfire Risk Threshold 0.40



Notification Preferences

- ☒ Email Alerts
- ☒ Push Notifications
- ☐ SMS Alerts
- ☒ Critical Alerts Only

Profile Settings

John Doe

john@email.com

Phone Number

Default Location

Karachi, Sindh

This location will be used as default for all monitoring dashboards.

Admin

OVERVIEW

Dashboard

User Management

API Monitor

SYSTEM

Model Management

Alert Config

Audit Logs

QUICK ACTIONS

User View

Logout

Admin Dashboard

System overview and health monitoring

System Online

1,247
Registered Users

8.5K
API Calls Today

99.9%
System Uptime

v2.4.1
Model Version

API Usage (Last 24 Hours)



API Health Status

NASA APIs Operational
EONET, POWER, FIRMS - All active

USGS Earthquake Operational
Response time: 245ms avg

OpenWeather Operational
Response time: 180ms avg

Recent Registrations

View All

USER	EMAIL	ROLE	JOINED	STATUS
AS Ahmed Shah	ahmed@email.com	User	2 hours ago	Active
FK Fatima Khan	fatima@email.com	User	5 hours ago	Active
HR Hamid Raza	hamid@email.com	Admin	1 day ago	Active

ML Model Status

Flood LSTM Model Active
Accuracy: 94.2% | Last trained: 2 days ago

Earthquake Predictor Active
Accuracy: 89.7% | Last trained: 5 days ago

Fire Detection CNN Updating
Progress: 75% | ETA: 15 mins

Manage Models

Recent System Logs

View All

[2025-04-16 14:32:15] [INFO] NASA EONET API fetched 12 new events
[2025-04-16 14:30:42] [INFO] Earthquake prediction model updated for Islamabad
[2025-04-16 14:28:10] [WARN] USGS API response time > 500ms
[2025-04-16 14:25:33] [INFO] User #1247 (Ahmed Shah) registered successfully
[2025-04-16 14:20:00] [INFO] Scheduled data sync completed - 847 records

OVERVIEW

Dashboard

User Management

API Monitor

User Management

Manage registered users and their permissions

Search users...

Add User

USER	EMAIL	ROLE	JOINED	LAST ACTIVE	STATUS	ACTIONS
AS Ahmed Shah	ahmed@email.com	User	2025-04-16	2 hours ago	Active	Edit
FK Fatima Khan	fatima@email.com	User	2025-04-15	5 hours ago	Active	Edit
HR Hamid Raza	hamid@email.com	Admin	2025-04-10	1 hour ago	Active	Edit

OVERVIEW

- Dashboard
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API Monitor

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- Model Management
- Alert Config
- System Logs

API Monitor

Real-time API health and performance monitoring

98.2%
Uptime

362ms
Avg Latency

USGS Earthquake

Online

Seismic Data

Latency: 333ms
Requests: 2,998/hr

NASA EONET

Online

Wildfire Events

Latency: 577ms
Requests: 3,237/hr

OpenWeather

Slow

Weather Data

Latency: 1373ms
Requests: 2,697/hr

GDACS

Online

Global Disasters

Latency: 649ms
Requests: 1,041/hr

API Response Time (Last 24 Hours)



Request Distribution



Recent API Calls

Time	Endpoint	Method	Status	Latency
19:35:32	/api/heatmap/data	GET	404	858ms
19:32:32	/api/heatmap/data	GET	200	290ms
19:29:32	/api/heatmap/data	GET	200	1660ms
19:26:32	/api/flood/predict	GET	200	120ms
19:23:32	/api/wildfire/active	GET	200	488ms
19:20:32	/api/alerts/all	GET	200	1661ms
19:17:32	/api/risk/assessment	GET	200	231ms
19:14:32	/api/earthquake/recent	GET	404	348ms
19:11:32	/api/flood/predict	GET	200	571ms
19:08:32	/api/flood/predict	GET	200	1794ms

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Model Management

AI/ML models for disaster prediction and risk assessment

5

Active Models

92.1%

Avg Accuracy



Flood LSTM

v2.3.1

Active

Accuracy

93.7%

Precision

85.4%

Recall

92.4%

Test

Export



Quake CNN

v1.8.2

Active

Accuracy

95.5%

Precision

92.3%

Recall

93.3%

Test

Export



Fire Detect

v3.0.0

Active

Accuracy

87.1%

Precision

83.7%

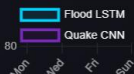
Recall

90.4%

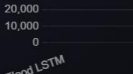
Test

Export

Model Accuracy Trend (7 Days)



Model Usage



Recent Training Jobs

+ New Training

Job ID	Model	Status	Progress	Started	ETA
JOB-5638	Flood LSTM	Failed	19%	17/04/2026, 19:39:26	-
JOB-1406	Risk Ensemble	Completed	23%	17/04/2026, 17:39:26	-
JOB-7751	Fire Detect	Failed	71%	17/04/2026, 15:39:26	-
JOB-1268	Flood LSTM	Queued	39%	17/04/2026, 13:39:26	-
JOB-8614	Risk Ensemble	Completed	52%	17/04/2026, 11:39:26	-
JOB-1161	Quake CNN	Completed	6%	17/04/2026, 09:39:26	-

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AI/ML models for disaster prediction and risk assessment

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92.1%

Avg Accuracy



Flood LSTM

v2.3.1

Active

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93.7%

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85.4%

Recall

92.4%

Test

Export



Quake CNN

v1.8.2

Active

Accuracy

95.5%

Precision

92.3%

Recall

93.3%

Test

Export



Fire Detect

v3.0.0

Active

Accuracy

87.1%

Precision

83.7%

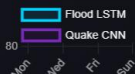
Recall

90.4%

Test

Export

Model Accuracy Trend (7 Days)



Model Usage



Recent Training Jobs

+ New Training

Job ID	Model	Status	Progress	Started	ETA
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JOB-1406	Risk Ensemble	Completed	23%	17/04/2026, 17:39:26	-
JOB-7751	Fire Detect	Failed	71%	17/04/2026, 15:39:26	-
JOB-1268	Flood LSTM	Queued	39%	17/04/2026, 13:39:26	-
JOB-8614	Risk Ensemble	Completed	52%	17/04/2026, 11:39:26	-
JOB-1161	Quake CNN	Completed	6%	17/04/2026, 09:39:26	-

OVERVIEW

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System Logs

Application logs and error tracking

```
[2026-04-17T14:41:14.676Z] [INFO] Data sync completed
[2026-04-17T14:40:44.677Z] [INFO] Earthquake data updated from USGS
[2026-04-17T14:40:14.677Z] [DEBUG] Earthquake data updated from USGS
[2026-04-17T14:39:44.677Z] [INFO] Flood risk threshold exceeded
[2026-04-17T14:39:14.677Z] [INFO] API request completed successfully
[2026-04-17T14:38:44.677Z] [DEBUG] Data sync completed
[2026-04-17T14:38:14.677Z] [DEBUG] Earthquake data updated from USGS
[2026-04-17T14:37:44.677Z] [INFO] Earthquake data updated from USGS
[2026-04-17T14:37:14.677Z] [DEBUG] User authentication successful
[2026-04-17T14:36:44.677Z] [INFO] Cache refreshed
[2026-04-17T14:36:14.677Z] [WARN] Data sync completed
[2026-04-17T14:35:44.677Z] [INFO] Cache refreshed
[2026-04-17T14:35:14.677Z] [WARN] Model prediction completed
[2026-04-17T14:34:44.677Z] [DEBUG] Background job started
[2026-04-17T14:34:14.677Z] [INFO] Flood risk threshold exceeded
[2026-04-17T14:33:44.677Z] [ERROR] Database connection established
[2026-04-17T14:33:14.677Z] [WARN] Background job started
[2026-04-17T14:32:44.677Z] [WARN] API request completed successfully
[2026-04-17T14:32:14.677Z] [DEBUG] Cache refreshed
[2026-04-17T14:31:44.678Z] [DEBUG] Cache refreshed
```